VANISHING OF THE FIRST COHOMOLOGY ON THE FIBRE MAKES THE STRUCTURAL MAP INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 21263-21343, immediately following the statement at 21234-21261.

Editorial note. Six cross-chapter pointers (Algebra, Coherent, Limits) are recorded in assembly.external_references as navigation pointers.

Editorial note. The final reduction to local rings is displayed with the explicit localised map, and the injectivity criterion is applied inside the span.

Allowed standard prerequisites (editorial).

- The flatness statement for H1 vanishing on the fibre. chapter lemma lemma-h1-fibre-zero-h0-flat, cited at proof line 21330. Supplies flatness of the direct image module at the relevant point.
- The residue field form of the vanishing criterion. chapter lemma lemma-h1-fibre-zero-h0-kappa, cited in the proof. Converts the fibre computation over the residue field into the corresponding statement for the local ring.
- A module injection criterion. Algebra, lemma algebra-lemma-mod-injective, cited at proof line 21340. Concludes that the map on local rings is injective from flatness plus injectivity of the mod-I reduction.
- Cohomology of proper pushforward and limit arguments. Coherent, proposition coherent-proposition-proper-pushforward-coherent and the Limits lemmas cited in the proof. Make the direct image coherent and allow the situation to be approximated by finitely presented data.

Source reference. Original reference 00ME: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02O5: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E7K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21193–21193. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E7J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21059–21059. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AI: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081F: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01Z6: focused-seeds/geometry/S9-morph-h1-fibre-zero-isom/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

of morphisms of schemes. Let $s \in S$ be a point. Assume

(1) *$X \to S$ is locally of finite presentation and flat at points of X_s,*
(2) *$Y \to S$ is locally of finite presentation,*
(3) *f is proper,*
(4) *the fibres of $f_s : X_s \to Y_s$ have dimension ≤ 1 and $R^1 f_{s,*}\mathcal{O}_{X_s} = 0$,*
(5) *$\mathcal{O}_{Y_s} \to f_{s,*}\mathcal{O}_{X_s}$ is an isomorphism.*

Then there is an open $Y_s \subset V \subset Y$ such that (a) V is flat over S, (b) $f^{-1}(V)$ is flat over S, (c) $\dim(X_y) \leq 1$ for $y \in V$, (d) $R^1 f_* \mathcal{O}_X|_V = 0$, (e) $\mathcal{O}_V \to f_*\mathcal{O}_X|_V$ is an isomorphism, and (a) – (e) remain true after base change of $f^{-1}(V) \to V$ by any $S' \to S$.

Proof. Let $y \in Y_s$. We may always replace Y by an open neighbourhood of y. Thus we may assume Y and S affine. We may also assume that X is flat over S, $\dim(X_y) \leq 1$ for $y \in Y$, $R^1 f_* \mathcal{O}_X = 0$ universally, and that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective, see Lemma 0E7J. (We won't use all of this.)

Assume S and Y affine. Write $S = \lim S_i$ as a cofiltered of affine Noetherian schemes S_i. By Limits, Lemma 01ZM there exists an element $0 \in I$ and a diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{f_0} & Y_0 \\ & \searrow \quad \swarrow & \\ & S_0 & \end{array}$$

of finite type morphisms of schemes whose base change to S is the diagram of the lemma. After increasing 0 we may assume Y_0 is affine and $X_0 \to S_0$ proper, see Limits, Lemmas 081F and 01Z6. Let $s_0 \in S_0$ be the image of s. As Y_s is affine, we see that $R^1 f_{s,*}\mathcal{O}_{X_s} = 0$ is equivalent to $H^1(X_s, \mathcal{O}_{X_s}) = 0$. Since X_s is the base change of X_{0,s_0} by the faithfully flat map $\kappa(s_0) \to \kappa(s)$ we see that $H^1(X_{0,s_0}, \mathcal{O}_{X_{0,s_0}}) = 0$ and hence $R^1 f_{0,*}\mathcal{O}_{X_{0,s_0}} = 0$. Similarly, as $\mathcal{O}_{Y_s} \to f_{s,*}\mathcal{O}_{X_s}$ is an isomorphism, so is $\mathcal{O}_{Y_{0,s_0}} \to f_{0,*}\mathcal{O}_{X_{0,s_0}}$. Since the dimensions of the fibres of $X_s \to Y_s$ are at most 1, the same is true for the morphism $X_{0,s_0} \to Y_{0,s_0}$. Finally, since $X \to S$ is flat, after increasing 0 we may assume X_0 is flat over S_0, see Limits, Lemma 04AI. Thus it suffices to prove the lemma for $X_0 \to Y_0 \to S_0$ and the point s_0.

Combining the reduction arguments above we reduce to the case where S and Y affine, S Noetherian, the fibres of f have dimension ≤ 1, and $R^1 f_*\mathcal{O}_X = 0$ universally. Let $y \in Y_s$ be a point. Claim:

$$\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y$$

is an isomorphism. The claim implies the lemma. Namely, since $f_*\mathcal{O}_X$ is coherent (Cohomology of Schemes, Proposition 02O5) the claim means we can replace Y by an open neighbourhood of y and obtain an isomorphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Then we conclude that Y is flat over S by Lemma 0E7K. Finally, the isomorphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$ remains an isomorphism after any base change $S' \to S$ by the final statement of Lemma 0E7K.

Proof of the claim. We already know that $\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y$ is surjective (Lemma 0E7J) and that $(f_*\mathcal{O}_X)_y$ is $\mathcal{O}_{S,s}$-flat (Lemma 0E7K) and that the induced map

$$\mathcal{O}_{Y_s,y} = \mathcal{O}_{Y,y}/\mathfrak{m}_s\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y/\mathfrak{m}_s(f_*\mathcal{O}_X)_y \to (f_{s,*}\mathcal{O}_{X_s})_y$$

is injective by the assumption in the lemma. Then it follows from Algebra, Lemma 00ME that $\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y$ is injective as desired. □